



World Class Heat pumps proudly Made in India



Proudly Designed & Made in India

Characteristics:

- 1) Adopts Tube in Shell heat exchanger and with inbuilt Circulating Pump inside.
- 2) Efficient Rotary compressor; Safe, Reliably, Stable running & durable.
- 3) Efficient Saginomya four-way valve; It is very efficient in defrosting in low ambient temp
- 4) Saginomya / SANHUA /EMERSON Electronic Expansion Valve (EEV) or Thermostatic expansion valve
- 5) Tube in tube heat exchanger inside with higher heat exchange efficiency than tube in shell heat exchanger used by others, with higher reliability & longer lifetime.
- 6) Green & Environment-friendly Refrigerant : R410A, R407c, R134A
- 7) Full consideration for noise control; Compressor rubber feet specially selected to reduce vibration.
- 8) Unique super energy saving controlling system can save extra 10% energy than other units in market.
- 9) Reliable Design and Strict Quality Control; All products 100% are tested and run at the factory to assure proper operation of all components and safety switches.

Core components

SCROLL COMPRESSOR  Highest durable and reliable compressor in the market EMERSON Copeland Panasonic Fully protected (High/low temp. High/low P; over current, phase protection) Anti liquid impact.	4 WAY REVERSING VALVE  Design ensures instantaneous changover with minimal pressure drop. The valves are equipped with a mechanism that prevents incomplete changover Danfoss SSG-MIV The 4-way reversing valves are UL and CE approved SANHUA	HYDROPHILIC FINNED EVAPORATOR  The hydrophilic coated finned evaporator with enhanced frosting and corrosion resistance. The corrugated louvered fin extend heat exchanging surface, thereby facilitating more contact with inlet air and improved efficiency of the evaporator.	TITANIUM HEAT EXCHANGER  High Efficiency of Heat Exchanger Using high efficiency finned tube, heat exchange heating area is 3.6 times than the smooth tube. Fins inside & outside tube creates turbulent flow of freon and water, which increases heating capacity. The compact helix structure of the coil ensures sufficient heat transfer between gas and water.
ELECTRONIC EXPANSION VALVE  Effectively controls the temperature & regulates optimum refrigerant flow for best in class C.O.P. Danfoss SSG-MIV SANHUA Better operating conditions means less faults and consequently a reduction in maintenance costs.	REFRIGERANT  Various option of Refrigerant liquid as per application that is environmentally friendly, non flammable and non toxic. R410A R417A R134A R407C	LARGE AIR FLOW FAN  Wide fan blade, low noise, high temperature and abrasion resistant. Large air flow design, means more air into the system, accelerating the heat exchange and improve efficiency.	INTELLIGENT LCD CONTROLLER  One touch intelligent controller with Touch screen design, clear working condition displaying, and self-check function of error.

MAIN FUNCTIONS OF HEAT PUMPS

Heat pump protection	High pressure protection	When water flow is too small or water temp. too high ,will prevent compressor to be damaged
	Low pressure protection	When refrigerant leaked , prevent compressor work without pressure.
	Compressor over current protection	Prevent compressor to work when current over max value
	Phase protection	Wrong phase/ absent phase protection (for 380V)
	Anti-freezing protection	Prevent water pipes and water tank freezing
Project function	Water level indication	Indicate high/low water level
	constant water temp. replenishing	Replenishing water temp. can be set, make water tank temp. constant
	Timing water temp. replenishing	Can set heat pump replenish cold water at certain time
	Timing hot water supply	Can supply hot water at certain time
	Cycle water(return water) setting	Can set cycle water temp.
	Timing start/stop heat pump	Can set 2 group of start/stop timing
	Manual switch	Separated start/stop signal can be connected with solr system(for choice



Option of Heating and cooling Model and WIFI models with App

Note : For selective models only, Get in touch with us for further details



Nomenclature :
V - Vindsol
CHP - Commercial Heat pump
Model No : 1720
H - Heating only
WF : with wifi and App
HC : with Cooling option

TECHNICAL SPECIFICATION		
Model	Units	VCHP - 1720H
Heating capacity	kW	20
	Btu/h	65877
Rated heated water output	Litre/hr	474
Rated outlet water temp.	°C	55
Max outlet water temp.	°C	60
Power	V/Ph/Hz	380 ~415V/3PH/ 50Hz
Rated input power	kW	4.5
Rated input current	A	9
Compressor type		Scroll*1
Compressor brand		Copeland
Throttle type		Electronic expansion valve (EEV)
Fan type		Low noise high efficiency axial type
Fan direction		Vertical
Fan No.		1
Fan Input power	W	250
Fan Speed	RPM	880
Evaporator fin type		Hydrophilic aluminium
Evaporator tube type		Innergroove tube
Heat exchanger ype		High efficiency tube in tube heat exchanger
Heat exchanger No.		1
Controller type		Intelligent & LCD
Ambient Temperature	°C	(-10°C~+45 °C)
Refrigerant type		R407c
Low pressure protection		Yes
High pressure protection		Yes
Automatic defroting system		Yes
Overload protection		Yes
Water pipe size	(mm)	Rc1(DN25)
Water flow volume	m3/h	4
Cabinet		Stainless steel
Unit Size	mm	855/775/1095
N.W	kg	165
Qty		2
Rate of each heat pump		182000
Total		364000
Testing condition : Heating: Ambient temp. (DB/WB): 25°C/20°C, water temp. (input/output): 20°C/55°C.The above data is only for reference: specific data is subject to the product		

HOT WATER Storage Tank Technical Specification and data sheet

Model		
Water tank volume lts	5000	
Insulation material	75mm thick with mineral wool, wire binding	
Cladding material	SWG 24 gauge aluminum cladding	
Shell / Dish end thickness	6/6 mm	
Inner tank coating	Enamelled Anti corrosion coat	
Inner tank material	MS	
Water conection	25 mm	
Sensor conection	1/2"	
Design working pressure	5 kgf	
Testing pressure	8 kgf	
Electrical backup connection	1 1/4".	
Qty - Nos	2	
Tank with cladding and insulation Price each	225000	Total 450000

Safety Kit Pack for Pressurised Tank			Rate	Qty	Amount
	Vacuum Relief Valve GST @ 18% extra	WATTS imported 3/4 inch lead free Vacuum Relief Valve automatically allows air to enter into the piping system to prevent vacuum conditions that could siphon water from the system and damage water equipment.	2800	2	5600
	Safety Pressure Relief Valve GST @ 18% extra	Safety Pressure Relief Valve 5 Bar , Protects hot water cylinders and heating components from high pressure			
	Automatic Air Vent valve GST @ 18% extra	Automatic Air Vent valve male thread. It automatically removes trapped air from a central heating installation.			
	hot water recirculation pumps (1WS)	Hot water Return Circulation pump pump , 100 % stainless steel multi stage pump , 4 m3/hr @ 15 m hear Flow rate - 8 m3/hr @ 15 m head. Make :Mechlion skid moubted system with 1W confuguration The 1W per heat pump	14500	2	29000
Total					34600

Consolidated Offer

1	Rate of each heat pump	364000
2	Tank with cladding and insulation Price each	450000
3	Accessories and Applicable Pumping system	34600
Total		848600
Gst @ 18%		152748
Grand total		1001348

TERMS & CONDITIONS

Prices applicable are Ex.Godown .F.O.R, Transportation charges extra at actual will be to customer account.

Packing and forwarding cost 2 % extra

GST @ 18 % Extra of order value

Payment to be made 50% as advance along with order and balance against proforma prior to delivery.

Delivery : within 4-5 weeks from date of confirmed order and PO

The rates are only for the Heat pump / Tank / accessories mentioned in the offer and not for any other items and accessories or plumbing with insualtion.

Installation testing and commissionign will be in dealer scope

plumbing works along with insualtion and Electrical work like incoming power supply cabling to heat pumps under customer scope.

Any civil work required at site will be in your scope for tank & heat pump positioning



Complete water Heating Solutions

A unit of Mechzephyr Engineering

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Domestic water heat pumps

Commercial Water Heat Pumps

Next Gen Thermodynamics Heat Pumps

Swimming pool heat pump